When More Search Can Worsen Deployment: Best-of-N Selection on an Incomplete Proxy

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Abstract

Can wider search worsen deployment when its selection score omits a cost? Selection overfitting and the effect of trading costs are established concerns. We use standard finite-pool rank probabilities to separate changes in winner exposure from the measured gross-to-net outcome gap. In a trading simulation, four pools of 500–1,500 ridge-regression strategies are ranked by gross Sharpe, ignoring execution costs. Wider search selects higher turnover, and the capacity component of the gap has a positive fitted slope on the primary search grid in all 16 pool-by-capital cells. From one to 32 trials, mean net Sharpe falls in 11 of 16 cells and rises in all four pool-A cells. The capacity effect is not statistically established: its joint selection–evaluation bootstrap interval for pool B at USD 500 million is $[-0.022, 0.121]$. Mean candidate gross Sharpe is negative in every pool, and all tested rules except an infeasible rule that selects on evaluation outcomes have negative mean net Sharpe on the comparison grid. Selecting on modeled net Sharpe improves that mean by 0.430, with paired interval $[0.192, 0.723]$, but leaves it at -0.071. The calculation makes search-induced exposure to an omitted term explicit. It also separates that measurement problem from selection noise: statistical correction cannot recover a cost the score does not measure.

1 What can wider search miss?

A researcher tests N candidates and selects the one with the highest score. This is *best-of-N selection*. Testing more candidates cannot reduce the best score seen. It can still worsen the outcome after selection when the score omits a relevant cost. A correction for noise in the measured score cannot recover a candidate-specific cost that was not measured.

We study this distinction in a trading simulation. A candidate is one configuration of a forecasting and portfolio-construction procedure. Its selection score is the *gross Sharpe ratio*, measured before execution costs. The deployment outcome is modeled net Sharpe, after those costs. A *proxy* is a score used in place of the full outcome of interest. Wider search favors better proxy ranks. Whether those ranks also have larger omitted gaps is an empirical question.

The strategies have no demonstrated trading edge. In each of four pools, the average candidate has negative evaluation-period Sharpe even before costs. Every tested non-oracle rule has negative mean net Sharpe on the comparison grid. An oracle here means an infeasible rule that selects using evaluation-period outcomes. We use these pools to study selection behavior, not to propose a profitable strategy.

The contribution is a paired measurement design and its empirical accounting. Standard finite-pool rank probabilities give the expected outcome of every search size without repeated search simulation. Comparing gross and net outcomes for the same selected candidates shows how much the omitted gap grows with search. The fitted gross-to-net gap slopes are positive in all 16 pool-by-capital

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cells. However, the joint selection–evaluation interval for the capacity component includes zero. Cost-aware selection improves the mean while leaving it negative. We report both results and the observed reversals.

Validation accuracy that omits serving latency, or a reward score that omits tool-use cost, motivates the same question. We test only the trading case. The rank calculation applies more widely under its sampling assumptions; the empirical direction does not transfer without measurements.

2 Relation to prior work

Random search and the dependence of validation performance on search effort are established topics [Bergstra and Bengio, 2012, Dodge et al., 2019, 2021]. Selecting among noisy estimates can create optimism and selection overfitting [Smith and Winkler, 2006, Cawley and Talbot, 2010]. More hyperparameter tuning can also reduce test performance [Schneider et al., 2025]. Order statistics describe ranked observations [David and Nagaraja, 2003]. The winner-rank law used below is standard for sampling without replacement [O'Neill, 2023]. Recent work uses related rank probabilities for best-of-N policy evaluation [Schweisthal et al., 2026]. Our question concerns a term absent from the selection score. It is complementary to estimating the measured score more accurately.

Relation to finance. Trading costs can remove much of the apparent return of high-turnover strategies [Lesmond et al., 2004, Korajczyk and Sadka, 2004, Novy-Marx and Velikov, 2016, Chen and Velikov, 2023]. Established remedies include portfolio rules that account for future trading costs [Gârleanu and Pedersen, 2013] and cost-aware combinations of firm characteristics [DeMiguel et al., 2020]. Closest to this study, Detzel et al. [2023] show that transaction costs change factor-model comparisons. Ignoring those costs can favor factors that are expensive to trade. Institutional cost evidence also differs from many academic cost assumptions [Frazzini et al., 2018]; we therefore report a lower-impact sensitivity.

A separate literature addresses data snooping and backtest selection. It includes the reality check and superior-predictive-ability test [White, 2000, Hansen, 2005], multiple-testing adjustments [Harvey and Liu, 2015, Harvey et al., 2016], the deflated Sharpe ratio and probability of backtest overfitting [Bailey and López de Prado, 2014, Bailey et al., 2017], and research protocols [Arnott et al., 2019]. These methods address uncertainty in measured returns. They cannot supply an unmeasured execution term. Our narrower contribution accounts for how search size changes exposure to that term in a fixed pool. We do not claim that the direction should surprise finance practitioners or that existing statistical corrections are ineffective.

3 Exact accounting for a fixed pool

Rank m candidates from best to worst by proxy score, with rank 1 best. Fix the order across search sizes, including a rule for exact ties. Draw N distinct candidates uniformly. Thus every subset of size N is equally likely. Let K_N be the winning rank. Write g_i for candidate i's gross outcome, u_i for its net outcome, and $q_i = g_i - u_i$ for their signed gap. Larger outcomes are better. A difference between Sharpe ratios is a score gap, not a dollar cost: costs can change both the mean and the standard deviation of returns.

Lemma 1 (Rank exposure; standard)**.** *For $1 \le N \le m$, the winning probabilities and tail probabilities are*

$$p_N(i) = \frac{\binom{m-i}{N-1}}{\binom{m}{N}}, \qquad T_N(k) = \Pr(K_N > k) = \frac{\binom{m-k}{N}}{\binom{m}{N}}. \tag{1}$$

For any fixed candidate gap values $q_1, \dots, q_m$, their expected selected value is

$$Q(N) := \mathbb{E}[q_{K_N}] = \sum_{i=1}^{m} p_N(i) q_i = q_1 + \sum_{k=1}^{m-1} (q_{k+1} - q_k) T_N(k). \tag{2}$$

Consequently, for $N < B \le m$,

$$Q(B) - Q(N) = \sum_{k=1}^{m-1} (q_{k+1} - q_k)\{T_B(k) - T_N(k)\}, \tag{3}$$

and each change in braces is nonpositive.

Proof. Rank i wins when it is drawn with $N-1$ worse ranks. The event $K_N > k$ requires all draws to miss the best k candidates. These counts give Equation (1), where a binomial coefficient is zero outside its combinatorial support. Write $q_i = q_1 + \sum_{k<i}(q_{k+1} - q_k)$ and exchange the finite sums to obtain Equation (2). Subtract the two budgets. A larger subset is less likely to miss the best k ranks, so $T_B(k) \le T_N(k)$. □

This is standard order-statistic bookkeeping, not a new theorem. If the gap never increases as rank gets worse, wider search cannot decrease its expectation. A nonmonotone gap has no universal direction. In the study, the direction comes from the recorded outcomes. Let $V_g(N) = \sum_i p_N(i) g_i$ and $V(N) = \sum_i p_N(i) u_i$. Since $V(N) = V_g(N) - Q(N)$, mean net performance falls precisely when the gap increase exceeds the gross improvement.

Why a proxy-only correction cannot identify the omission. Hold two candidates' observed proxy data fixed. Their omitted terms can be swapped so that a different candidate is best in each of two compatible cases. Any rule using only the proxy data sees the same input in both. It cannot guarantee the best deployment choice in both cases, even if it randomizes. This elementary identification remark concerns an unrestricted missing term. Statistical corrections still address noise in the measured score.

What we measure. For $N \in \{1, 2, 4, 8, 16, 32\}$, compute the selected gross and net means with Equation (1). Fit each six-point curve against $\log_2 N$ by ordinary least squares. Its slope summarizes change per doubling on this grid. It need not describe each doubling separately. The gross slope minus the net slope is the slope of $Q(N)$. We call this the *search-induced gap*, denoted Γ in Supporting Methods. The search grid remains below ten percent of each pool; conclusions about the final few ranks require a separate comparison.

4 Trading case study

4.1 Candidates and execution model

The study uses daily data for 258 survivor-filtered US equities through 24 July 2026. Selection uses 2010–2018; evaluation uses 2019–July 2026. Four seeded pools contain 500 configurations each in A and D and 1,500 each in B and C, drawn from 24,000 combinations within each pool's feature-set realization. Each fits ridge regression, which penalizes large fitted coefficients. Configurations vary in forecast horizon, training window, regularization, smoothing, and portfolio construction. Models are refitted every 21 trading days. Training excludes forward-return labels that cross the fitting boundary. Predictions become long and short target holdings with zero net dollar exposure and unit gross exposure, the sum of absolute holdings.

The selector maximizes selection-period gross Sharpe. Sharpe is $\sqrt{252}$ times mean daily modeled return divided by its daily standard deviation. The simulator sets dollar targets from a fixed capital amount, marks existing holdings to market, and then trades toward the targets. Each stock's daily trade is capped at 10% of trailing 21-day median daily dollar volume. Each executed dollar incurs a spread of 1.5 basis points (0.00015), plus impact $0.5\sigma\sqrt{p}$, where σ is daily volatility and p is participation in that volume. These are assumed execution charges, not measured fills [Almgren and Chriss, 2001, Almgren et al., 2005, Tóth et al., 2011].

Signals formed through one close affect the next decision boundary and earn the following close-to-close return. Daily profit and charges are normalized by gross exposure actually held. Partial fills can leave net exposure. Financing, borrowing charges, margin constraints, and wealth-based deleveraging are absent. The output is a constant-notional execution simulation, not a funded investor-account return. Supporting Methods Sections 1–3 specify the features, eligibility rules, timing, and holdings equations.

4.2 Performance levels and search slopes

The mean candidate's gross evaluation Sharpe is -0.095, -0.109, -0.114, and -0.131 in pools A–D. At $N = 32$, gross-score selection gives gross Sharpe $+0.039$, -0.054, -0.002, and -0.036.

Search improves these means, but the selected gross levels remain near zero. At USD 500 million, mean net Sharpe changes from -0.581 to -0.537 in A, -0.610 to -0.820 in B, -0.650 to -0.718 in C, and -0.661 to -0.804 in D when search expands from one to 32 trials.

Outcome and capital	Pool A	Pool B	Pool C	Pool D
Gross	$+0.0288$	$+0.0153$	$+0.0250$	$+0.0216$
Net, USD 25 million	$+0.0213$	-0.0057	$+0.0110$	$+0.0007$
Net, USD 100 million	$+0.0164$	-0.0186	$+0.0027$	-0.0100
Net, USD 500 million	$+0.0113$	-0.0342	-0.0080	-0.0232
Net, USD 2 billion	$+0.0157$	-0.0387	-0.0124	-0.0239

Table 1: Evaluation Sharpe slopes per doubling of search, using selection-period gross rankings and the six-point grid. Net slopes are below gross slopes in all 16 pool-by-capital cells, but only nine net slopes are negative. A positive slope does not imply a positive performance level. The USD 2 billion case is strongly affected by the trade cap.

The search-induced gap is positive in all 16 primary cells (Table 1). This does not mean that net performance always falls. Nine net slopes are negative. Comparing only the endpoints, net Sharpe falls in 11 of 16 cells. It improves at every capital in pool A and at USD 25 million in pool C. All cells share one market history.

To separate the fixed spread from capacity effects, evaluate four paths in a fixed order: uncapped gross, capped gross, capped with spread, and capped with spread and impact. Adjacent differences in Sharpe slopes sum to the total search-induced gap. The capping and impact terms together form the *capacity component*, denoted Γ_{cap} in Supporting Methods. This term measures the part associated with limited trading volume and size-dependent charges. It ranges from 0.0172 to 0.0460 per doubling at USD 500 million. The decomposition depends on the path order; it is not a unique attribution of cash costs.

4.3 Uncertainty in selection changes the conclusion

At USD 500 million, pool B has the largest estimated capacity component, 0.0460, and is the reference bootstrap case. When selection and evaluation periods are resampled jointly and ranks are recomputed, its 95% percentile interval at USD 500 million is $[-0.0219, 0.1209]$; 88% of replicates are positive. Thus the capacity component is not statistically established under selection uncertainty. The gross slope's joint interval also includes zero, $[-0.0438, 0.0678]$.

These stationary block bootstraps resample consecutive groups of days, with 1,000 replicates and mean block length 21. Selection and evaluation windows are resampled separately. Within each window, all candidates use the same day indices. Holding the ranking fixed and resampling only evaluation gives the narrower capacity-component interval $[0.0383, 0.0551]$. It answers a different question, conditional on the observed ranking. Joint intervals also include zero at the other retained block lengths and at USD 25 and 100 million. Supporting Methods Section 4 gives the full uncertainty specification.

4.4 Mechanism and observed reversals

Turnover is executed traded notional divided by held gross exposure. Expected selected turnover rises with search in each primary cell. At USD 500 million in pool D, it rises from 0.137 with one trial to 0.199 with 32. Regressing the capacity-related signed gap on turnover and multiplying by the fitted selected-turnover slope accounts for 0.89–1.20 of the primary capacity component at USD 500 million. This is descriptive accounting; the relationship is less stable in shorter windows. Figure 1 shows the rank reweighting, candidate gaps, and cancellation between local contributions.

The same rank calculation exposes contrary cases (Table 2). A gap need not rise at every step just because its full-grid slope is positive. At USD 500 million, the late grid $\{8, 16, 32\}$ gives a capacity-component slope of -0.0070 in pool A. The expected gap is not monotone across the primary search budgets in pool A's four capital cells. The middle liquidity tercile also reverses the effect.

The primary impact coefficient is an assumption. Matching the cited temporary-impact form at 10% participation gives a square-root coefficient near 0.11. Under that lower coefficient, the USD 500

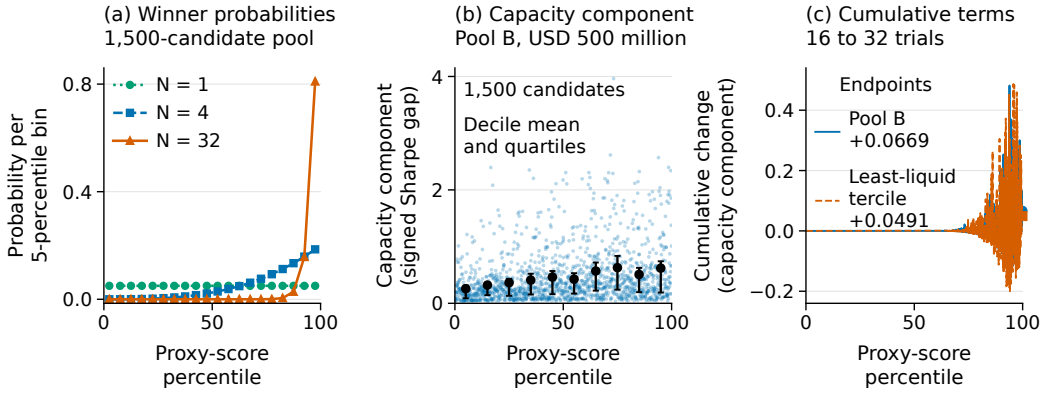


Figure 1: Search reweights an observed, nonmonotone gap. (a) Exact winner-rank probabilities in five-percentile bins for a 1,500-candidate pool. Higher proxy-score percentiles are better. (b) Pool B at USD 500 million: the capacity component of the signed gross-minus-net Sharpe gap for each candidate. Black marks show rank-decile means and interquartile ranges, which describe candidate spread, not uncertainty in a population mean. (c) Cumulative terms in Equation (3) for 16 to 32 trials at USD 500 million. Terms accumulate from the lowest proxy-score percentile; this reverses the best-first order in the equation but preserves the sum. Positive and negative terms offset, leaving changes $+0.0669$ in pool B and $+0.0491$ in the least-liquid tercile. These are fixed-pool calculations.

Check	Variation	Capacity component at USD 500 million	Negative cases
Search grids	Five grids anchored at one trial; 64 eligible pool-by-grid-by-capital cells	$[0.0172, 0.0571]$	None in anchored grids; A is -0.0070 on the late grid
Execution	Trade cap, spread, and impact assumptions	$[0.0069, 0.1106]$	None on this grid
Lower impact	Coefficient 0.11 instead of 0.5	$[0.0052, 0.0106]$	None on this grid
Walk-forward	Six time periods	$[-0.0096, 0.2418]$	Seven of 72 cells across three capitals
Liquidity	Three retrospective terciles	Low 0.0968; middle -0.0279; high 0.0334	Middle tercile reverses at all three capitals

Table 2: Sensitivity ranges and reversals. These settings reuse market data and candidates; cell counts do not represent independent confirmations. Walk-forward selection windows are disjoint, but earlier evaluation dates can enter later selection windows. Supporting Methods Section 4 explains the checks. The complete capital-specific rows are in `finance_additional_checks.json`.

million capacity component remains positive but is roughly three to four times smaller. Its size is therefore sensitive to execution calibration.

Removing the single worst evaluation configuration in each cell preserves the primary signs and changes the capacity component by at most 8.6%. This does not bound every relative change: pool D's near-zero net slope at USD 25 million changes by about 89%, from approximately 0.0007 to 0.0001. Capital ordering also has a limit. The capacity component increases from USD 25 million through USD 500 million in each pool, but pool A falls from 0.0172 at USD 500 million to 0.0130 at USD 2 billion. More capital does not imply a larger signed Sharpe gap in every case.

5 Changing the selection score

We retain the best-of-N rule and change what it scores. A net-score rule uses selection-period modeled net Sharpe at an assumed capital. A calibrated turnover penalty subtracts turnover times the pool's mean selection-period gross-to-net gap divided by mean turnover. Both require the execution simulator. Table 3 averages four pools, three assumed capitals, and three evaluation capitals at $N = 32$. These 36 dependent cells use USD 25, 100, and 500 million.

Selection score	Mean net Sharpe	Gain over gross
Gross	-0.501	0.000
Turnover decile screen	-0.298	$+0.203$
Simulator-free penalty, $\eta_{\mathrm{ref}} = 0.5$	-0.169	$+0.332$
Simulator-free penalty, $\eta_{\mathrm{ref}} = 0.11$	-0.229	$+0.272$
Calibrated turnover penalty	-0.048	$+0.453$
Net at assumed capital	-0.071	$+0.430$
Deflated net at assumed capital	-0.071	$+0.430$
Ex-post oracle	$+0.220$	$+0.721$

Table 3: Changing the selection score improves mean net Sharpe, but every non-oracle mean remains negative. Each mean equally weights 36 cells at 32 trials, across four pools, three assumed capitals, and three evaluation capitals. The oracle uses evaluation outcomes and is infeasible at selection. Simulator-free penalties use full-panel universe summaries, which can include evaluation-period dates. These dependent cells do not establish future profitability.

The turnover decile screen removes the highest tenth of selection-period turnover before gross selection. The simulator-free penalty subtracts a cost-model coefficient times uncapped turnover; η_{ref} sets its assumed impact coefficient. The deflated-net rule uses the Deflated Sharpe Ratio to adjust for multiple trials and non-normal returns. Supporting Methods Section 4.1 gives each rule's formula and inputs.

Net-score selection improves mean net Sharpe by 0.430 relative to gross selection. A paired joint block-bootstrap interval for the gain is $[0.192, 0.723]$. The calibrated turnover penalty gives a gain of 0.453 with interval $[0.198, 0.728]$. Each uses 1,000 replicates with mean block length 21, sharing day indices across candidates, pools, and selectors and recomputing scores and selections. Both gains are positive in all replicates. Their resulting mean levels remain negative. These are post hoc simulator comparisons, not independently validated selection rules.

The net-score and deflated-net-score criteria have nearly identical aggregate mean performance here. Nearly equal aggregate means do not show that the two rules rank candidates identically. The result does not show that deflated Sharpe is generally ineffective. Statistical adjustment and measurement of omitted costs solve different problems.

6 Limitations and transfer

The strategies have no demonstrated edge: mean candidate gross Sharpe is negative in every pool, and every tested non-oracle rule has negative mean net Sharpe on the comparison grid. The positive capacity components share one market history, and the reference joint interval for that component includes zero. Costs are modeled rather than measured from fills; the effect is smaller under lower impact. The universe is survivor-filtered, with retrospective eligibility and missing failed or delisted

firms. Selection-score comparisons are post hoc. The execution model omits financing and several account constraints. Its order-dependent Sharpe decomposition is not a unique cash-cost attribution. Exact rank weights do not remove these empirical limits.

What would transfer. Lemma 1 applies to any fixed-pool best-of-N search with uniform sampling without replacement and a fixed score order. The sign still depends on the outcomes attached to those ranks. Adaptive candidate generation, stopping based on observed scores, or other search laws require different probabilities. Validation loss that omits serving latency and a reward score that omits tool-use cost have the same omission structure, but we have not tested them. A test needs a stated deployment objective, the omitted term measured for each candidate, and evidence that its relation to proxy rank survives uncertainty in the selection data.

The practical lesson is to measure the outcome that selection is meant to improve. In these pools, the gross-to-net gap has a positive fitted search slope, and cost-aware selection improves mean net Sharpe. Neither result establishes profitability or a stable effect on a new market path. A correction for selection noise cannot supply a cost absent from the measured objective.

Data, funding, and assistance

The public repository at `https://github.com/dvgyl/when-more-search-can-worse n-deployment` contains this paper, Supporting Methods, figure data, retained finance aggregates, and verification scripts. Vendor price data are not redistributed. The package contains the retained aggregates behind the reported values, a script that checks the principal values, and the script that renders Figure 1. It does not replay the raw market study. Supporting Methods identifies the original sources and reconstruction limits.

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